Talk with your latent - can LLM do scientific reasoning directly from data?

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Abstract

Recent advances in Large Language Models (LLMs) offer unique opportunities for scientific tasks, yet their ability to reason over complex numerical data remains largely unexplored. While several works investigate the ability of LLMs to create textual descriptions of scientific objects (e.g., proteins), we ask whether they can understand those objects and answer comparative questions about them. This kind of reasoning beyond description is crucial for the development of useful co-scientists. We test two different astrophysical datasets, involving both stars and galaxies, with pre-trained multimodal features that serve as the LLM data tokens. Adding those tokens as additional information, we fine-tune an 8B Llama model on various scientific reasoning tasks. We find that while the model can describe the stellar objects reasonably well, they fail when asked about populations, outlier detections, and cross-modal correlations. We further show that by training a small non-language head that would highlight the scientific information (e.g., a flow matching model for population questions), and adding its latent features as an extra token, the performance of the LLM on the relevant tasks improves dramatically. The results are consistent across two independent datasets and highlight the lack of statistical reasoning abilities in current language-based models and the need for dedicated models to transform the data in a way that helps LLMs understand it.

1 Introduction

The integration of Large Language Models (LLMs) into the scientific workflow marks a paradigm shift in how researchers interface with information. While these models have demonstrated remarkable proficiency in some scientific tasks, their utility remains largely confined to the textual and visual domains. A critical frontier in "AI for Science" is the transition from processing scientific text to reasoning directly over the physical data that defines scientific discovery. Standard LLM architectures struggle with numerical reasoning [1], in part because subword tokenization does not preserve the magnitude and continuity of numerical values [2], creating a gap between natural language and empirical data.

The main motivation of this work is the need to combine physics foundation models and LLMs in a scientifically useful manner. Each has distinct advantages, and the potential for knowledge alignment between them has far-reaching outcomes. One is interpretability: an LLM that operates over both natural language and learned scientific representations could serve as a 'bridge' between humans and machine learning representations, which are usually hard to interpret. A second is scientific discovery: foundation models trained on large and diverse datasets encapsulate latent information beyond their downstream tasks, and that information can support non-trivial population analysis or complex cross-modal correlations. Unfolding it, however, requires analysing high-dimensional

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features directly, whereas the common practice of projecting them to a low-dimensional summary discards exactly the structure such analyses depend on. Most current uses of LLMs with scientific data take the form of textual descriptions, or of agents that delegate the analysis to dedicated models and read off their results. Whether an LLM can perform scientific reasoning *directly* from learned data representations is therefore still an open question, and it is the question we study here.

Contribution. We connect frozen survey encoders to an 8B language model through a trained projector, adapting the language model itself with either LoRA or full fine-tuning, and evaluate it on two independent astrophysical datasets (Kepler-field stars, COSMOS-Web galaxies) across per-object and pairwise questions, and on a more challenging question for each: the odd member of a set of stars, and the agreement between two views of a single galaxy. *First*, we find a sharp asymmetry. Reading learned tokens, the model describes single objects and compares pairs mostly correctly on both datasets, and judges whether a galaxy's shape and star-formation history agree for 97–99$\%$ of galaxies. It fails when the answer requires comparing a whole group: it names the odd star and its property in a third of groups, falling from 0.51 with four stars to 0.15 with 32. A linear probe ranks the odd star as well as the catalog values do, so the signal is in the features and the failure is in using it. Supplying the catalog values as text does not help: the model either names most of the group or, when it reasons first, does no better than the model reading the tokens, and a larger general-purpose model (Qwen3.5-27B) does no better either. *Second*, performing the group computation outside the model and returning it as a single extra token doubles the stellar score ($0.33 \rightarrow 0.66$), keeps it flat across group sizes, and holds on groups whose odd star was never the odd one in training. This narrator was also trained differently (LoRA instead of full fine-tuning), and the matching control has not been run (Section 5). On galaxies, where the narrator already succeeds, the head token adds nothing. *Third*, verified templated prose yields a substantially more factual model than fluent LLM-written prose on the same task, because the LLM-written text states more than the input determines.

2 Related Work

LLMs as scientific assistants. Advances in LLMs have already had a wide-ranging impact on how we do science. LLM agents can now review and write scientific papers [3, 4], act as a 'virtual lab' that suggests and implements novel research ideas [5], or perform equation discovery [6]. However, these agents typically access scientific data indirectly, by writing code to an external tool (e.g., ESM, AlphaFold-multimer), an approach formalised in the tool-augmentation literature [7, 8]. Recently there has been growing interest in predicting physical properties of proteins with LLMs [9, 10]. That line is close to ours, but asks a different question: property prediction and textual description from sequential data, rather than comparative and population analysis over continuous data.

Scientific foundation models and scientific LLMs. Foundation models have become popular across scientific fields and show strong performance in many domains [e.g., 11–18]. Notably, these models do not treat natural language as a modality, so a clear separation persists between scientific foundation models and LLMs. The term scientific-LLM usually refers instead to injecting scientific knowledge into existing LLMs [for reviews, see 19, 20], most often by fine-tuning on high-quality text and images to improve reasoning [21–23], object description [24], search [25], and chat [26]. Such models outperform general LLMs on domain tasks [27], but remain confined to text and images.

Connecting frozen encoders to frozen language models. The architecture we use is standard outside the sciences: a trained projector maps a frozen encoder's outputs into the embedding space of a frozen language model, which is adapted with a small number of trainable parameters [28–31]. This recipe is well-established for producing descriptions of a single input. Whether it supports comparison across several inputs, which is what population-level scientific questions require, has received little attention.

3 Data and Model

Terminology. We use six names throughout. The *narrator* is the 8B language model we train: it reads an object's data tokens and answers questions about the object. The *head* is a small model, not a language model, trained separately to compute one quantity that a question depends on (Section 4);

its output enters the narrator as one extra token, the *head token*. The *templater* is the deterministic program that writes our training text from catalog facts, and the *writer* (DeepSeek-V4) is the LLM we use in its place in one comparison. Generated text is scored by the *verifier*, a program that checks every stated fact against the catalog, and by the *judge* (Qwen3.5-27B), an LLM that reads each description against the ground truth.

We test two different astrophysical datasets, representing different domains and physical properties. The first dataset is a *stellar* multimodal dataset comprising 190,091 stars, observed by 5 different modalities coming from different telescopes - *Kepler*[32], TESS[33], LAMOST[34], APOGEE[35], and Gaia[36]. Each modality is processed through a specific encoder, and the modalities are aligned with a model that factorizes the latent space into shared and specific subspaces. The result is one shared feature and 5 per-modality features, each projected to the token embedding space of the LLM as its own small group of tokens. The second dataset is galaxies from the COSMOS-Web survey [37], including two modalities: images and star-formation history (SFH). Similar to the stellar dataset, we use a pretrained model that aligns the two modalities (here using CLIP [38]) and project the frozen features into the LLM's token embedding. Notably, the two datasets are different in many aspects: number and type of modalities, pretrained models, and scientific questions. Using the two datasets in the same pipeline highlights general vs data-specific conclusions.

training prose. One of the challenges in this pipeline is creating the training dataset. This should include ground-truth prose per object. Contrary to other fields, in astrophysics data is usually stored only in numerical catalogs, so one needs to create the prose from scratch. Our main method is a *template-based* method: a deterministic program that picks a subset of facts from the catalog, chooses a phrasing from a bank of templates, and writes prose. Every sentence is checked against the record by the verifier before it enters the corpus. The check is per-fact: it guarantees that the facts a description states are true of the object, not that the description mentions every unusual object in a group. Both datasets share two types of prose: description of a single object (single), and comparison between pairs (pair). We also test deeper reasoning capabilities with two additional tasks. In the stellar dataset, we use population prose: a group of N objects with a description of the typical members, and a callout of the ones that do not belong and on which property. In the galaxy dataset, the task is to judge whether a galaxy's shape agrees with its star-formation history, a relation between its two modalities. In Section 4 we compare the template-based method with a second method: an *LLM-based* generation, where the writer produces the population prose from the same catalog facts, and every claim is verified.

Objects, facts and descriptions. The stellar set has 190,091 stars, of which we evaluate on a held-out set of 19,010. Because every star is described in many training records and the stellar training text was not built split by split, 99.2% of these evaluation stars appear in *some* training description; Section 4.3 tests whether this matters. The galaxy set has 137,696 galaxies, split $80/10/10$ into training, validation and test galaxies, and all galaxy training text is written about training galaxies only, so the test galaxies are never seen in training.

here we might want to emphesize that the datasets representes two different paradigms - the stellar dataset treat every prose as a unique sample, regardless of which star it came from. this enables larger diversity (?) but might cause memorization of specific samples, which we test in the experiments (4.3 and 4.1, which show that some descriptions of the stellar case are recall from training. The other is a fixed split wihhc we do in the galaxy case. This might justify the stellar split and memorization results. if we do it, we should also mention this difference in the previous paragraph.

Each star's catalog record holds 13 numbers ($T_{\rm eff}$, $\log g$, [Fe/H], mass, radius, age, luminosity, distance, parallax, Gaia's astrometric quality index RUWE, apparent and absolute brightness, and rotation period) and categorical facts: evolutionary state (dwarf, subgiant, giant); how the star compares with the rest of the sample (e.g., among the most metal-rich quarter, or hotter, older or faster-rotating than most); its Galactic disk population and α-element abundance; its activity level; and flags for binarity, planet hosts, flares and open-cluster membership. Each galaxy's record holds stellar mass, star-formation rate and state (quiescent to starburst), redshift, Sérsic index (how centrally concentrated its light is), visual morphology and disturbance, formation age, and the shape of its star-formation history (rising or falling, recent activity, quenching, formation time).

The templater writes each description by sampling, independently, which facts to state (a random three to six), a style, and a framing. For stars it also adds, each with probability one half, a chemistry block (disk population, α-abundance) and an activity block (chromospheric activity, variability amplitude). Stars have three styles (natural prose, observation log, bullet list) and four framings (plain description, outlier flag, comparison with other stars, comparison with the Sun). Galaxies have four styles (technical paper, observation log, teaching text, concise summary), four framings (plain description, comparison with the galaxy population, comparison with the star-forming main sequence, outlier flag) and three lengths. On stars, each fact is then worded by drawing from a bank of alternative phrasings (917 alternatives at 141 points in the text); the galaxy templater has one wording per fact and style, so its variety comes from the facts, style, framing and length. A

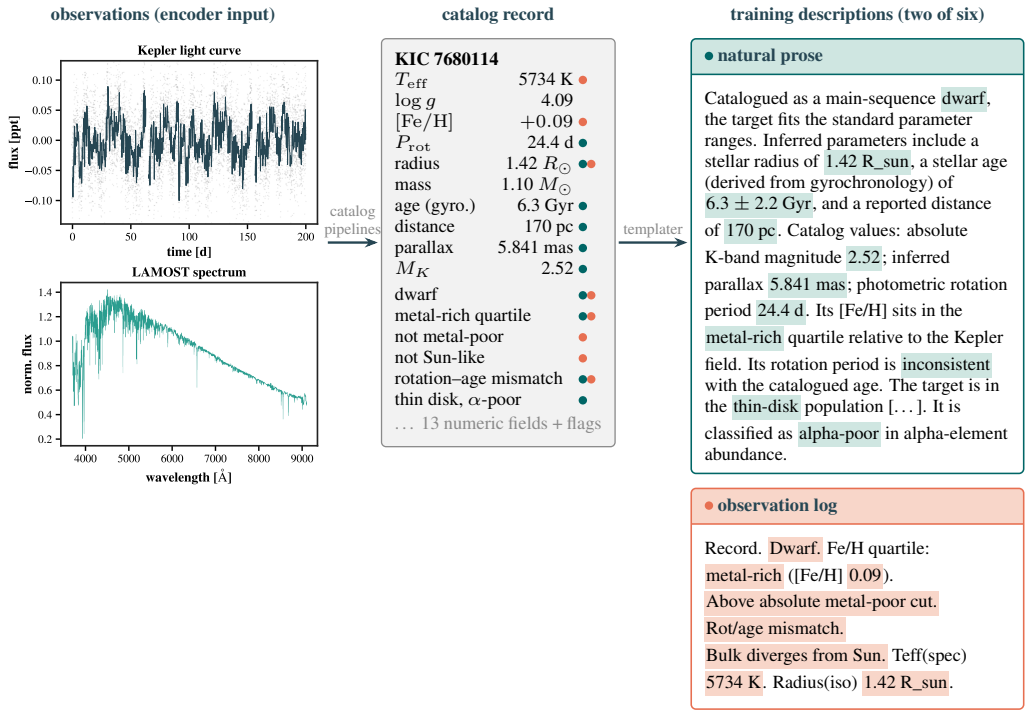


Figure 1: **From observations to training text**, for one star (KIC 7680114). *Left*: two of its observations, the Kepler light curve (6-hour bins over 30-minute data) and the LAMOST spectrum. The frozen encoders read these; the language model never sees them directly. *Middle*: its catalog record (abridged). *Right*: two of the six training descriptions written about it, verbatim, in two styles. Each draws a different random subset of facts. Dots mark which facts each description uses.

star is described about five times, each time with different facts and words (Figure 1). The result is highly diverse: every stellar single and pair description is unique, and 99.9% remain unique with all numbers masked; for galaxies, 99.9% of single descriptions are unique and 70% remain so with numbers masked. In total, the stellar corpus has 1.93M records (1.01M single, 0.69M pair, 0.23M population); each population record is a group of 4 to 32 stars with an odd member on one of six properties. The galaxy corpus has 288k records (90k single, 198k pair). The galaxy agreement task uses a separate, deliberately narrow corpus of 45,817 records (three phrasings per case, also from training galaxies only) in which galaxies whose morphology and star-formation history disagree are oversampled to 33%.

Training recipe. A narrator trained on templated text can learn to reproduce the templates instead of reading the data. We counter this with two changes to the input. First, we vary its format: an object's data tokens appear alone, after a question or a short passage, separated from the description by intervening text, or among the tokens of several other objects together with a request to describe one of them. Second, part of the intervening text consists of real sentences from astrophysics papers, filtered so that none states a relation between properties we later test. The two are needed together (Section A.3). Varying the format alone improves accuracy but makes the narrator copy training descriptions almost verbatim; adding the paper sentences removes that copying, and accuracy improves further. Population records are always presented as the data tokens alone. The recipe was chosen on the galaxy data and reused unchanged for stars.

all the memorization tests should be somehow unified in terms of the narrative - we test them with the div+arxix pipeline but also with the different types of datasets - independent prose vs. sample-based split. can we somehow draw a clear picture from everything together? eventhogh they are done on different datasets?

4 Results

4.1 singles and pair questions

We first test the single and pair questions, comparing two types of Llama-8B models: Llama-3.1 and AstroSage [27], a Llama-3.1 model that was fine-tuned specifically on arXiv astrophysics pa-

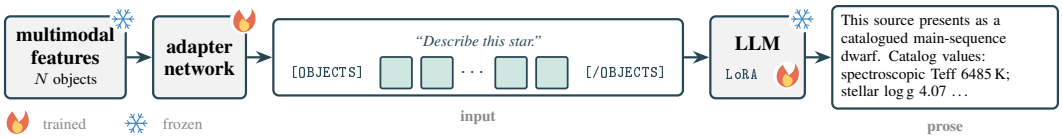


Figure 2: **The pipeline.** An adapter projects each object's multimodal features into tokens; the model processes them, outputs prose, and the loss is taken on the prose alone. The language model is adapted either by full fine-tuning or with LoRA.

pers. Table 1 shows the comparison on both datasets. Single and pair descriptions are scored by the verifier, which extracts every numeric and categorical claim from the generated text and checks it against the catalog: *numeric accuracy* is the fraction of stated values within a per-quantity tolerance of the catalog value, and *qualitative* (single) or *relational* (pair) accuracy is the fraction of class words or comparisons that match it (tolerances and definitions in Section A.4). Both models perform well on both tasks, and the differences between them are small: AstroSage is ahead in five of the eight cells, and no difference exceeds 0.04. We use AstroSage for the rest of the paper. Section A.5 breaks the accuracy down by property: class statements are mostly right, while some stated numbers (age, parallax, galaxy stellar mass) are often off by more than 10%. Probes trained on the same tokens (Section A.6) show that on galaxies the narrator states about what the tokens contain, while on stars part of its numeric precision is recall of catalog values from the training text.

the comparison with probes should include more information - both models are better than linear probe (right?) and the comparison with non-linear probe (MLP) shows the recall claim. we should also emphesize the advantage of using language model for descriptions - we needed to train seperate probes per property. this is more complex

4.2 scientific reasoning tasks

Single and pair questions consists of description and simple comparisons and requires almost no reasoning beyond the basic properties. It is also similar to the protein description tasks as in [10], and therefore it is not surprising that an LLM performs well on such questions. Next, we test the performance on questions that requires some form of scientific reasoning. As mentioned in Section 3, those are finding the odd member of a group of stars, and judging whether a galaxy's shape agrees with its star-formation history. The rest of the paper focuses on those tasks. Table 2 shows the results of the two reasoning tasks. Figure 4 shows one example generated description for each task. The two tasks behave differently. On stars, the narrator alone, reading only the data tokens, fails: it names the odd member and its property in only a third of the groups. On galaxies, the narrator alone succeeds: trained with LoRA, it gives the right verdict for 97–99% of galaxies. Interestingly, full fine-tuning performs slightly worse than LoRA, especially on quiescent galaxies.

Before adding anything to the stellar narrator, we check whether its failure is a failure of information. A cross-fit linear probe on the same aligned features, asked to rank the members of a group, matches what the true catalog values achieve (rank-1 0.660 from the shared subspace and 0.686 from the full feature, against 0.660 for the catalog itself; Section A). The signal the task needs is present in the features the model reads, so the failure is in the readout and not in the representation. The last column of Table 2 adds the head token (see Section 3). In the stellar case the head is a flow matching model that was trained to learn the joint distribution of the aligned features and catalog labels, conditioned on the rest of the group. In the galaxy case, it is a cross-modal predictor trained to predict one modality from the other. On stars, the head narrator doubles the score ($0.33 \rightarrow 0.66$). On galaxies, it is no better than a narrator trained the same way without the token (Section A.2). The stellar narrator without a head was fully fine-tuned while the head narrator was trained with LoRA, and the matching stellar control (LoRA without a head) has not been run (Section 5).

run the missing experiment

The asymmetry in performance, together with the results in Table 1, points to the number of objects in the question as the important factor for successful generation of the *narrator*. The galaxy verdict concerns a single object and follows from two of its own properties, its shape and whether it still forms stars, which the narrator can read from that galaxy's tokens. Even though morphology-SFH disagreement is deeper than mere description, the *narrator* still performs almost perfectly. In the stellar case, finding the odd star requires comparing every member with the rest of the group, a computation over the whole set that no single star's tokens contain. The narrator's strong performance on single and pair questions supports the hypothesis that population-related questions are inherently more challenging for the narrator. To test this, we vary only the number of stars while keeping the question fixed: the test groups contain between 4 and 32 stars, and Figure 3 shows the score

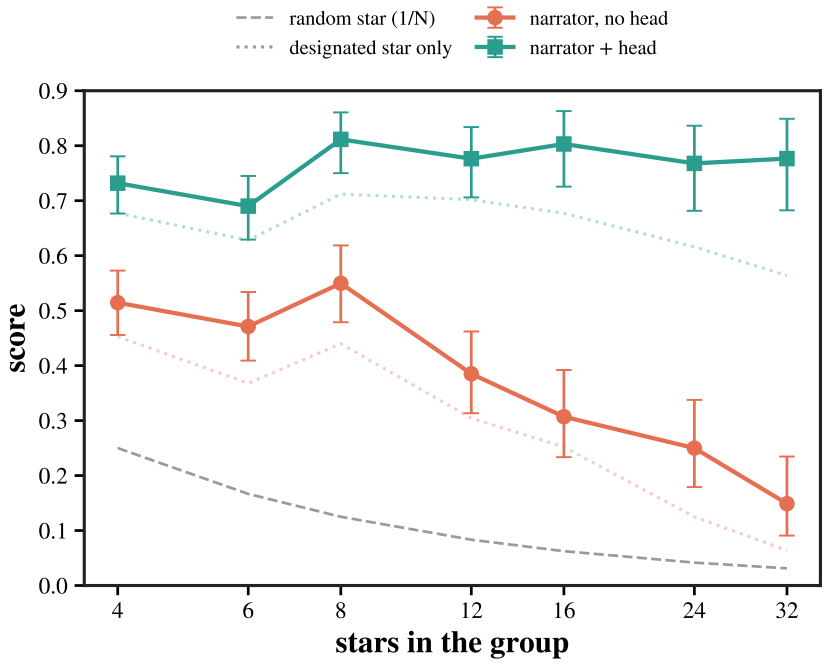


Figure 3: **The stellar task becomes harder with the number of stars.** Fraction of groups in which the narrator names a star that truly has the odd property, and the property, as a function of the number of stars in the group, for the narrator without and with the head (1,203 test groups with an odd member). Error bars are 95% intervals. Dotted lines: the same, counting only the designated odd star (the score of Table 2). Dashed line: the chance of picking the odd star at random ($1/N$); the score also requires the right property, so its chance level is lower still.

as a function of group size. Here we count a description as correct if it names *any* star that truly has the odd property, together with that property: the test groups designate one odd star, but larger groups more often contain a second star with the same property (24% of groups with 4 stars, 84% with 32), and naming it is a true statement. Without the head, the score falls steadily as the group grows, from 0.51 with 4 stars to 0.15 with 32 (Spearman $\rho = -0.21$, $p = 10^{-13}$). With the head, it stays between 0.69 and 0.81 at every size and does not decline. Counting only the designated star gives the same picture at lower values (dotted lines). The difficulty of the stellar task therefore grows with the number of objects the narrator must compare, and the head narrator does not suffer from it. Because the two narrators also differ in training (LoRA against full fine-tuning), part of this stability may come from the training rather than the head (Section 5). Since every star adds the same number of tokens, a larger group is also a longer input, and our data cannot separate the two. Next, we perform two ablation tests for the stellar population task: whether the improvement from using the head comes from reasoning or memorization, and whether handing the *narrator* the full catalog numbers prevents the failure.

4.3 is this reasoning, or memorization?

As described in Section 3, 99.2% of the stars we evaluate on appear in *some* training description. That is the same for every narrator, so the paired comparisons above are fair, but it leaves an obvious objection: the head narrator might be recalling which star was called unusual during training rather than working it out.

We test this directly, on the stellar track. From the test fold we keep only groups whose odd member was *never* an odd member anywhere in training: the star may have been described many times, but never as the one that stands out. Only 2.3% of candidate groups survive that filter, leaving 1,000. If the head narrator is recalling, it should lose most of its advantage here. Table 3 shows this experiment. The advantage does not shrink where memorisation is impossible; it grows. The head is

Galaxy: does its shape agree with its history?

Catalog. Late-type disk (P=0.75); quiescent ($\log \mathrm{sSFR} = -11.6$); declining star-formation history. A disk that has stopped forming stars: shape and history *disagree*.

Narrator (no head token) ✓

A quiescent late-type disk — a cross-modal outlier, because disks rarely quench while retaining their morphology. Its star-formation history is declining.

Shape, star formation and verdict all match the catalog.

Stars: which one does not belong?

star	class	T_{eff} [K]	$\log g$	R [$R_\odot$]
0	dwarf	5778	4.39	1.05
1	**giant**	4662	**2.45**	**11.14**
2	subgiant	5628	3.89	2.00
3	dwarf	5394	4.55	0.81

Narrator, no head ×

These 4 stars form a homogeneous population of field stars. No anomalies detected along the field-star axis.

Narrator + head ✓

Sample N=4: single stars. index 1: a giant [logg = 2.44 dex, R = 11.14 Rsun, logL = 1.72 log(Lsun), giant].

Figure 4: **Example generations on the two reasoning tasks**, verbatim and decoded greedily. *Left*: a quenched disk, whose shape and star-formation history disagree. The narrator trained without the head token states the shape, the star-formation state and the verdict correctly, as it does for 97–99% of test galaxies. *Right*: a test group of four stars containing one giant. The narrator without the head reports no odd member; the narrator with the head names the giant, with values that match the catalog.

Table 1: **Base model: Llama-3.1-8B vs. AstroSage-8B**, identical corpus, recipe and seed; rule-scored, greedy; the better model in bold. Metrics: *numeric acc.* = fraction of numbers the text states that are within 10% of the catalog value (Section A.4); *qualitative acc.* (single) = fraction of class words (dwarf/giant, disk/elliptical, ...) that match the catalog; *relational acc.* (pair) = fraction of comparisons ("hotter than", "more massive") that match.

metric	Llama-3.1-8B	AstroSage-8B
galaxies — one object		
numeric acc.	**0.750**	0.711
qualitative acc.	0.881	**0.896**
galaxies — pair		
numeric acc.	0.410	**0.444**
relational acc.	**0.795**	0.771
stars — one object		
numeric acc.	0.651	**0.660**
qualitative acc.	0.888	**0.891**
stars — pair		
numeric acc.	**0.776**	0.766
relational acc.	0.858	**0.883**

worth +0.469 on this set against +0.330 on the full test set, and the two narrators move in opposite directions on it: the narrator without the head falls (0.332 → 0.279) while the head narrator does not. That is the opposite of what a memorization explanation predicts.

4.4 giving the model the numbers as text

If the features carry the signal and the model cannot use it, an obvious alternative is to skip the features and hand the model the catalog numbers directly. We test this on the stellar task. Each group is shown as a table with one row per star, listing its catalog values (temperature, surface gravity, metallicity, radius, mass, and others) together with the two catalog flags the head also receives: whether the star is a binary and whether it belongs to an open cluster. With these flags the table holds everything the head is given, so the comparison is fair for all five kinds of odd star. The model sees two worked examples and is asked, in words, to name the star that does not belong and the property it differs in. We test two models: the base AstroSage-8B, without our training,

Table 2: **The reasoning tasks.** Each row is one task and scorer; the columns are narrators that differ in how they were trained and whether they read the head token. Stars: the fraction of the 1,203 test groups with an odd member in which the text names it *and* its property, scored by the verifier and by the judge, which agree on the size of the effect. Galaxies: the fraction of test galaxies with the correct agreement verdict (Section A.4), on the 9,609 star-forming and 1,068 quiescent galaxies. The stellar narrator trained with LoRA and no head token was not run (Section 5).

		narrator, no head token		narrator + head
task	scorer	full fine-tune	LoRA	LoRA
stars — name the odd member	verifier	0.245	—	**0.633**
and its property	judge	0.332	—	**0.662**
galaxies — shape vs. history,	star-forming	0.980	**0.988**	0.986
correct verdict	quiescent	0.884	0.974	**0.978**

Table 3: 1,000 stellar groups whose odd member was never an odd member in training, greedy, judge-scored. *score* is the fraction of descriptions that name the true odd member and its property; *named* counts naming the odd member regardless of property, and *property* naming the right property regardless of member.

	score	named	property
narrator	0.279	0.329	0.343
narrator + head	**0.748**	**0.782**	**0.761**
Δ	+0.469	+0.453	+0.418

and Qwen3.5-27B, a larger general-purpose model. Each is tested twice: answering directly, and reasoning step by step before answering. Table 4 compares them with the two narrators on the same 1,203 groups.

No model that reads the numbers comes close to the narrator with the head (0.662). AstroSage reading the table reaches the second-best score (0.473), but by naming most of the group (82% of the stars): naming the odd star among them is then no better than chance (0.92×). Asked to reason first, it names few stars and scores 0.288, below the narrator without the head. The larger model is more selective: it names the odd star at about the same rate as the head relative to the number of stars it names. However, it often gets the odd star's property wrong, and scores 0.341 answering directly and 0.324 after reasoning, no different from the narrator without the head (p=0.7 in both cases). In most groups (94%) the larger model was still reasoning when it reached the length limit of 4,096 tokens and was made to answer from what it had, so its reasoning result holds for this budget only. The gap to the head also holds for every group size (Section A.7). Reading the numbers lets a model describe the stars, but not reliably pick out the one that does not belong.

Table 4: **Giving the model the numbers as text**, on the 1,203 test groups with an odd star (all five kinds), greedy. The text-reading models see every catalog value plus the binary and open-cluster flags. *score* and *named* are as in Table 3; *stars named* is the fraction of the group the text names; *lift* divides *named* by it (1.0 = no better than naming that many stars at random). Every text-reading model scores below the narrator with the head ($p < 10^{-19}$, exact McNemar on *score*).

	score	named	lift	stars named
AstroSage-8B, reads the table	0.473	0.761	0.92×	82%
AstroSage-8B, reads the table, reasons first	0.288	0.447	2.31×	19%
Qwen3.5-27B, reads the table	0.341	0.512	3.69×	14%
Qwen3.5-27B, reads the table, reasons first	0.324	0.570	**4.13**×	14%
narrator	0.332	0.442	3.36×	13%
narrator + head	**0.662**	**0.797**	4.08×	20%

4.5 Templates vs. LLM prose

Everything above is trained on prose written by the deterministic templater. The obvious alternative is to let a capable LLM write it, which has the potential for better diversity and fluency. To compare the two, we built an LLM-based corpus: the writer (DeepSeek-V4) was given the same catalog facts and asked to write the group descriptions, and every claim it made was checked against the record before the text was accepted. The training text is, in that sense, true. Table 5 compares the two corpora with everything else held fixed: same base model, same prompt, and no head (both narrators read the data tokens only and are fully fine-tuned). We see that training on the writer's fluent prose makes the narrator invent: contradictions per group go from 1.9 to 7.2, and not one group out of $1{,}191$ is described fully correctly. The training text was verified true; the model trained on it is not. This is aligned with the finding of [39] who showed that template-based serialization outperforms LLM-based serialization for tabular data. The two corpora differ in how much they commit to. Per group description, the templater writes a median of 21 words containing 5 numbers and names 17% of the members; it states the population and the member that does not belong, and says nothing about the rest. The writer uses 82 words containing 17 numbers and names 83% of the members, giving most of them a full quantitative profile. Training on the second teaches the narrator to do the same. On the test groups, the narrator trained on writer prose writes a median of 78 words against 17, states 8 values per group against 1, and names 72% of the members against 26%. Most of its extra contradictions (70%) are such numeric claims, and most of those concern members *other* than the outlier, whose values the template-trained narrator rarely states at all. Per stated value, the two narrators are contradicted at a similar rate (0.52 vs. 0.46): the writer-trained narrator is wrong more often mainly because it says more. Values for the typical members are hard to read for both narrators: for either one about 40% of them contradict the record (verifier). A narrator trained to state them for every member therefore mostly adds errors. This is not the whole story: on the outlier itself, the writer-trained narrator's stated values contradict the record far more often (18% vs. 2% of values, verifier). Verification of the training text does not help in either case, because the problem is not that the text is false; it is that it asserts more than the input determines.

Table 5: **Which prose to train on.** AstroSage-8B, same prompt, latent tokens only (no head), full fine-tune; greedy, judge-scored, on the $1{,}191$ groups with an odd member judged for both narrators (paired; hence 0.333 here vs. 0.332 on the $1{,}203$ groups of Table 2). *score* is as in Table 2; *covered* = the text correctly describes the typical (non-odd) members; *contradictions/group* = the number of statements the judge marks as contradicting the record; *all correct* = score, covered, and no contradiction. Covered, contradictions and all correct differ at $p < 10^{-45}$, score at $p = 9{\times}10^{-6}$ (exact McNemar; Wilcoxon for contradictions).

training prose	score	covered	contradictions/group	all correct
template	**0.333**	**0.573**	**1.88**	**0.128**
writer (DeepSeek-V4), verified	0.258	0.178	7.16	0.000

5 Limitations

6 Conclusion

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A Appendix

A.1 Is the signal in the features? The ranking probe

The group task asks which member does not belong. We test whether that is answerable from the aligned features alone, without a language model in the loop. For each group we rank its members by one quantity and ask where a qualifying star lands. Two readouts are compared on exactly the same groups: the true catalog value, and a cross-fit ridge regression that predicts the same quantity

from the aligned feature (5-fold, so no star is scored by a model that saw it). If the probe tracks the catalog, the signal survives the alignment.

Two details matter. First, only the three continuous axes can be ranked by a scalar; binarity and cluster membership are categorical and are excluded (103 of 259 anomalies). Second, the composer designates one intruder but leaves the other members unconstrained on that axis, so a group often contains several stars carrying the same label — 60.5% of anomalies on the full fold, and 81.7% on dwarf/giant. Naming any of them is a true statement about the group, so a readout is credited for the best-placed qualifying star rather than only the designated one. "Extreme" is distance from the group median in units of its median absolute deviation, which does not reveal which direction the answer lies in.

Table 6: **Ranking the odd member from the features alone.** n=156 continuous-axis anomalies over 129 groups. *rank-1* = a qualifying star is the single most extreme member; *top-3* = one is among the three most extreme. The probe reads only what the language model reads.

readout	rank-1	top-3
catalog values (oracle)	0.660	**0.942**
probe, shared subspace (64-d)	0.660	0.840
probe, full feature (1024-d)	**0.686**	0.891
by axis, full feature vs. catalog		
dwarf / giant	0.911 *vs* 0.857	1.000 *vs* 0.929
hot / cool	0.755 *vs* 0.694	0.939 *vs* 0.959
metal rich / poor	0.373 *vs* 0.412	0.725 *vs* 0.941

The probe matches the catalog at rank-1 and keeps 89–95% of it at top-3. The information for this task is therefore in the features.

The catalog row is not a ceiling on the task. It bounds what a *single ranked quantity* can do, and it falls short of 1.0 because the label is a threshold test ("$[\mathrm{Fe/H}] > 0.3$") while a ranking is a distance test: a star can sit far from the group median without crossing the threshold. A readout that models the group jointly, the head of Table 2, is not bound by it.

Scoring the narrators by the same rule, crediting any equally-qualifying star, raises both narrators ($0.332 \rightarrow 0.419$ without the head, $0.662 \rightarrow 0.756$ with it) and leaves the head's lead intact: $+0.33 \rightarrow +0.34$ (n=1,203 groups, exact McNemar $p=7 \times 10^{-81}$).

A.2 The galaxy narrators: token or training?

The galaxy head narrator differs from the fully fine-tuned narrator in two ways: it reads the head token, and it is trained with LoRA on a frozen base. To separate the two, we trained a third narrator on the *same* text with the *same* LoRA settings and no head token, and we also ran the head narrator with its token scrambled at test time (each galaxy receives the head token of another test galaxy). All four narrators describe the same 10,677 test galaxies, decoded greedily, and are scored as in Section A.4.

Table 7: **Galaxy agreement task, four narrators.** Fraction of test galaxies with the correct verdict. p-values are exact McNemar tests on the same galaxies, against the LoRA narrator with the head token.

	star-forming (9,609)	quiescent (1,068)
full fine-tune, no head token	0.980 ($p=5\times10^{-6}$)	0.884 ($p=2\times10^{-23}$)
LoRA, no head token	**0.988** (p=0.002)	0.974 (p=0.50)
LoRA + head token	0.986	**0.978**
LoRA + head token, scrambled	0.978 ($p=3\times10^{-9}$)	0.972 (p=0.34)

Three things follow. First, the LoRA narrator without the head token is as good as the one with it: indistinguishable on quiescent galaxies and slightly better on star-forming ones. Second, the fully fine-tuned narrator is clearly worse on quiescent galaxies; most of its errors there come from stating that a quiescent galaxy still forms stars. Third, scrambling the head narrator's token costs it

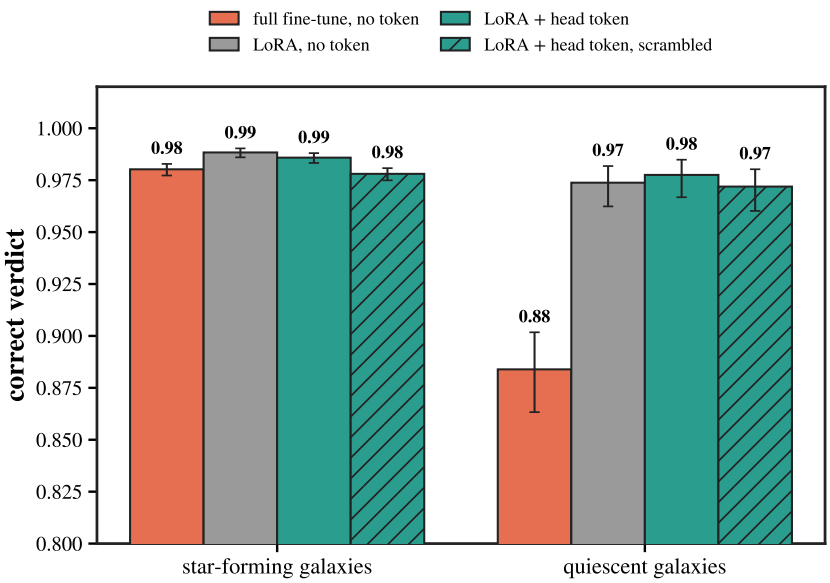


Figure 5: **Galaxy agreement task**, the numbers of Table 7. Error bars are 95% intervals.

a little on star-forming galaxies, so the narrator does read the token, but the information it carries is already available from the data tokens. On galaxies, the improvement over the fully fine-tuned narrator therefore comes from the training, not from the head token. The stellar equivalent of this control was not run (Section 5).

A.3 The training recipe

Table 8 tests the two input changes of Section 3 on the galaxy data with Llama-3.1-8B. *Novel phrasing* is the fraction of generated descriptions that do not copy a training description (Section A.4); low values mean the narrator recites the templates.

The upper block adds the components one at a time on a 95k-record subset, with everything else fixed. Varying the input format raises accuracy on single objects, but the narrator starts reciting its training text: novel phrasing falls from 0.40 to 0.18 on single objects and from 0.42 to 0.04 on pairs, and pair accuracy drops. Adding the paper sentences reverses this (novel phrasing 0.39 and 0.68) and gives the highest accuracy on single objects and on pair comparisons; only numeric accuracy on pairs stays below the baseline (0.384 vs. 0.416).

The lower block, on the full 288k-record corpus, adds the paper sentences *without* varying the format. On single objects both accuracies rise (numeric $0.540 \rightarrow 0.647$, qualitative $0.836 \rightarrow 0.870$), but copying increases (novel phrasing $0.15 \rightarrow 0.06$). Pairs do improve. This comparison differs from the upper block in corpus size, learning-rate schedule and the share of intervening text taken from papers, so we read it only as consistent with the claim that the paper sentences help when the input format is varied. A one-variable test of the paper sentences without varied formats is in progress.

A.4 Evaluation

Rule-based verifier (single and pair records). A deterministic program extracts every numeric and categorical claim from the generated text and checks it against the object's catalog record. No model is involved. It reads values written in running text ("a temperature of 5800 K") and in list form ("Surface gravity: 4.3"). A negated statement counts as a claim of the opposite ("not Sun-like" is correct for a star that is not Sun-like), and a galaxy's star-formation trend is checked over the time window the text names (e.g. "declining over the last 5%").

Table 8: **Training recipe**, galaxies, Llama-3.1-8B, greedy, scored by the verifier. *num.*: numeric accuracy; *qual./rel.*: qualitative (single) or relational (pair) accuracy; *novel*: novel phrasing. Upper block: cosine learning-rate schedule with label smoothing in every row. Lower block: the first two rows use a constant learning rate and no label smoothing; the last row is the full recipe used throughout the paper. Blocks are not comparable with each other.

	single object			pair		
	num.	qual.	novel	num.	rel.	novel
95k-record subset						
baseline	0.500	0.828	0.404	**0.416**	0.731	0.419
+ varied formats	0.519	0.843	0.180	0.340	0.695	0.043
+ varied formats + papers	**0.526**	**0.855**	0.386	0.384	**0.750**	**0.682**
full corpus (288k records)						
baseline	0.540	0.836	0.151	0.374	0.622	0.268
+ papers only	0.647	0.870	0.060	**0.422**	**0.806**	**0.631**
full recipe	**0.750**	**0.881**	**0.537**	0.410	0.795	0.355

- *Numeric accuracy*: the fraction of stated values within 10% of the catalog value. For quantities given as logarithms ($\log g$, $[\mathrm{Fe/H}]$, $\log M_\star$, $\log \mathrm{sSFR}$) this means 10% in the underlying quantity, i.e. within $\log_{10} 1.1 = 0.041$ dex. The galaxy star-formation trend index is signed and is zero for a flat history, so a relative tolerance is undefined for it; it keeps an absolute tolerance of 0.05. A value off by more than three times the tolerance counts as a *contradiction*. Table 9 gives the accuracy and the mean absolute error of each quantity separately.
- *Qualitative accuracy* (single): the fraction of class words that match the catalog. For stars: evolutionary state (dwarf, subgiant, giant) and group-relative classes (metallicity rank in four classes, hot/cool, fast/slow rotator, young/old, luminous/faint, Sun-like, metal-poor on an absolute scale). For galaxies: morphology class (early/late disk, elliptical, mixed), star-formation state (quiescent, green valley, star-forming, starburst) and mass class.
- *Relational accuracy* (pair): the fraction of comparisons that match the catalog. For stars: directional comparatives (hotter, more massive, larger, older, higher $\log g$, more luminous, more metal-rich, and their opposites). For galaxies: same or different morphology, star-formation-history trend, disturbance and density cluster.
- *Grounded* $= 1-$ the fraction of claims not supported by the record.
- *Novel phrasing* measures whether the narrator writes its own sentences or recites training text. For each generated description we find the most similar description in the training corpus (MinHash nearest neighbour over word 5-grams) and compute the fraction of the generation's 5-word sequences that appear verbatim in it. A generation *copies* a training description if at least half of its 5-word sequences do. Novel phrasing is $1-$ the fraction of generations that copy.

Scoring the galaxy task. We read three statements from each galaxy description: the galaxy's shape (disk or spheroid), whether it still forms stars, and the verdict, i.e., whether the description calls the two in agreement. Each is read with a short list of words. For the verdict, "mismatch", "disagree", "outlier" or "unusual" mark disagreement, and "concordant", "in accord", "concur" or "agree" mark agreement. In the training text these words are exact: every description containing a disagreement word belongs to a disagreeing galaxy, and every one containing an agreement word to an agreeing galaxy (Section A.4). We report the fraction of galaxies with the correct verdict, separately for star-forming and quiescent galaxies. The split matters because 90% of the test galaxies are star-forming, and the errors concentrate in the quiescent minority, where a narrator must notice that a galaxy has stopped forming stars. Whenever a narrator's verdict is right, its stated shape and star-formation state are also right in more than 99.9% of cases.

The judge (stellar population records). The verifier's extractors expect the template's phrasing, so the population task is also scored by the judge (Qwen3.5-27B), a text-only model that never sees any token. It reads each group description against the ground-truth record and is instructed to judge factual content only, not style. It returns: whether the text names the true odd member (*named*);

Table 9: **Numbers stated for single objects**, per quantity, by the Llama-3.1-8B and AstroSage-8B narrators of Table 1 (greedy). *acc.*: fraction within 10% (Section A.4); *MAE*: mean absolute error in the quantity's unit; n: number of stated values. Quantities stated fewer than 10 times by both models are omitted.

		Llama-3.1-8B			AstroSage-8B		
quantity	unit	n	acc.	MAE	n	acc.	MAE
stars							
$T_{\rm eff}$	K	331	0.99	67	293	1.00	67
$\log g$	dex	355	0.59	0.072	293	0.59	0.074
mass	$M_\odot$	342	0.85	0.050	222	0.83	0.064
radius	$R_\odot$	155	0.57	0.34	180	0.61	0.40
age	Gyr	284	0.34	1.46	178	0.31	1.50
parallax	mas	270	0.43	0.22	147	0.36	0.30
galaxies							
$\log M_\star$	dex	88	0.20	0.19	226	0.23	0.14
$\log \mathrm{sSFR}$	dex	229	0.30	0.12	195	0.27	0.14
redshift	—	739	0.85	0.066	543	0.86	0.045
Sérsic index	—	0	—	—	61	0.16	1.40
formation age	Gyr	82	0.55	0.054	324	0.58	0.17
SFH mean time	—	1692	0.89	0.012	2248	0.87	0.013
SFH trend index	—	1726	0.70	0.048	1890	0.66	0.051
recent mass fraction	—	419	0.65	0.026	1058	0.64	0.023

whether it attributes it to the correct property (*property*); whether the population statement describes the typical members (*covered*); and a list of claims that contradict the record (*contradictions*). We derive *score* = named *and* property, and *all correct* = score, covered, and no contradiction. Only groups that contain an odd member are scored. All narrators are judged on the same groups, so comparisons are paired (exact McNemar for yes/no outcomes, Wilcoxon for contradiction counts). The judge is given no numeric tolerance and is stricter than the verifier: of the numeric contradictions it lists that we could parse, 56–60% (depending on the narrator) are within the verifier's tolerance. Its contradiction counts are therefore an upper bound, and they inflate more for a narrator that states more numbers.

Galaxy agreement task. Each description is reduced to three statements, read with fixed word lists taken from the nine single-galaxy templates of the training text. *Shape*: "late-type disk" or "early-type disk" (disk), "elliptical" or "lenticular" (spheroid); general phrases that state the rule, such as "spheroidal systems", are ignored. *Star formation*: the first of "star-forming" or "still forming stars" (star-forming) and "quiescent" or "has quenched" (quiescent); the galaxy's own state always precedes the rule clause, as in "This disk is quiescent — … disks usually remain star-forming". *Verdict*: the words in Table 10; a description containing words from both lists, or from neither, has no verdict. Table 10 checks every word against the ground truth of the $36{,}654$ single-galaxy training descriptions; the shape and star-formation rules are likewise correct on 100% of them. A galaxy's verdict is scored correct if it matches the catalog. No narrator left a galaxy without a verdict.

Table 10: **Verdict words in the galaxy training text.** For each word, the number of the $36{,}654$ single-galaxy training descriptions that contain it, and the fraction of those that describe a galaxy whose shape and star-formation history truly disagree.

	word	descriptions	galaxy disagrees
disagreement	mismatch	4,012	100%
	disagree	3,961	100%
	outlier	4,123	100%
	unusual(ly)	6,893	100%
agreement	concordant	8,177	0%
	in accord	8,054	0%
	concur	8,327	0%
	agree	8,177	0%

A.5 Accuracy per property

Figure 6 breaks the single-object accuracy of AstroSage-8B down by property, on the whole test set (19,010 stars, 3,000 galaxies). Most class statements are right more than 90% of the time. The exceptions are informative. For stars, the metallicity rank is right in 68% of cases, and most errors confuse two neighbouring classes (below and above the median); subgiants are often called dwarfs (59% correct, against 98% for dwarfs and giants). Among stated numbers, temperature and mass are usually within 10%, while radius, surface gravity, parallax and age often are not; age is off by 1.5 Gyr on average. For galaxies, redshift and the mean star-formation time are accurate, while stellar mass, specific star-formation rate and the Sérsic index are rarely within 10% (stellar mass is off by 0.14 dex on average).

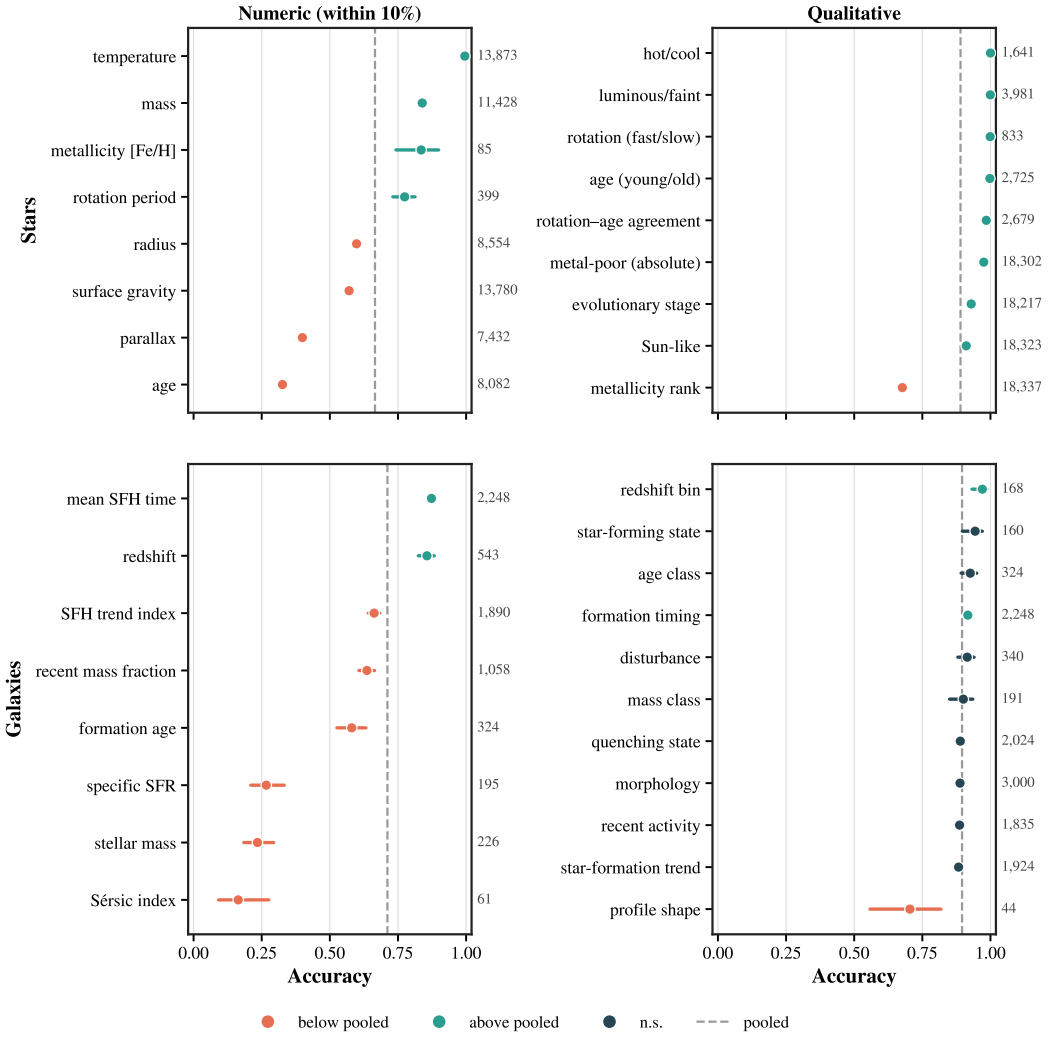


Figure 6: **Single-object accuracy per property**, AstroSage-8B, greedy, whole test set. Left: fraction of stated numbers within 10% of the catalog value (Section A.4); right: fraction of class statements that match the catalog. Bars are 95% intervals; the dashed line is the pooled accuracy; colour marks properties significantly below or above it; numbers at the right are how many statements were scored. Properties stated fewer than 30 times are omitted.

A.6 Does the narrator say what the tokens contain?

To see whether the narrator's errors come from the tokens or from the narrator, we fit two probes for every property on the same inputs the narrator receives (the aligned star tokens; the image and star-

formation-history embeddings of a galaxy). The first probe is linear (ridge or logistic regression); the second is non-linear with one hidden layer of 512 units. Both are trained on the training fold and tested on the test fold. For each property we compare them with the narrator on the objects where the narrator stated that property (Figure 7).

Galaxies. No test galaxy appears in the training text, so this is a clean comparison. The narrator says about as much as the non-linear probe can read. On classes they are equal for morphology, disturbance and the star-formation trend; elsewhere the probe is ahead by 0.03–0.07 (by 0.09 for the Sérsic shape, which the narrator states for 1% of galaxies). On numbers the probe is more accurate for the star-formation-history quantities and the specific star-formation rate, and the narrator is more accurate for redshift and stellar mass. The largest gap is not accuracy but coverage: the narrator states the stellar mass, specific star-formation rate, redshift or Sérsic index for only 2–18% of galaxies, although the probes read them well ($R^2 = 0.85$–0.99, except the Sérsic index, $R^2 = 0.43$).

Stars. The narrator is more accurate than both probes on most numbers, for example radius (0.40 against $0.82\,R_\odot$ for the non-linear probe). Part of this is recall. The narrator states the exact catalog value far more often than a probe can: temperature to the kelvin in 35% of its statements, against 0.5% for the non-linear probe. Almost every such value is written in some training description of that star, which exists for 99.2% of test stars (Section 3); when the value appears in none of them, exact matches drop to the probe's level. The effect on the average error is small: it is only slightly lower where the training text gave the number. On classes the narrator and the probes agree to within 0.02.

Weak in both. Some properties are hard for the probes too: star age ($R^2 = 0.55$), the metallicity rank (66%, against 25% by chance), and the galaxy Sérsic index. For these, the information is mostly missing from the tokens, and the narrator's errors (Figure 6) reflect the tokens rather than the narrator.

A.7 Reading the numbers, by group size

Table 11 breaks Table 4 down by the number of stars in the group. The narrator without the head falls from 0.45 to 0.06 as groups grow; the models that read the table fall less, but stay below the narrator with the head at every size.

Table 11: **Score by number of stars in the group**, for the models of Table 4 (same $1{,}203$ groups; n = groups of that size).

stars in the group	4	6	8	12	16	24	32
n	276	242	191	161	127	112	94
AstroSage-8B, reads the table	0.55	0.52	0.48	0.30	0.50	0.41	0.45
AstroSage-8B, reads the table, reasons first	0.29	0.29	0.33	0.26	0.28	0.33	0.22
Qwen3.5-27B, reads the table	0.40	0.39	0.31	0.37	0.26	0.24	0.30
Qwen3.5-27B, reads the table, reasons first	0.30	0.38	0.37	0.35	0.31	0.22	0.27
narrator	0.45	0.37	0.44	0.30	0.25	0.13	0.06
narrator + head	**0.68**	**0.63**	**0.71**	**0.70**	**0.68**	**0.62**	**0.56**

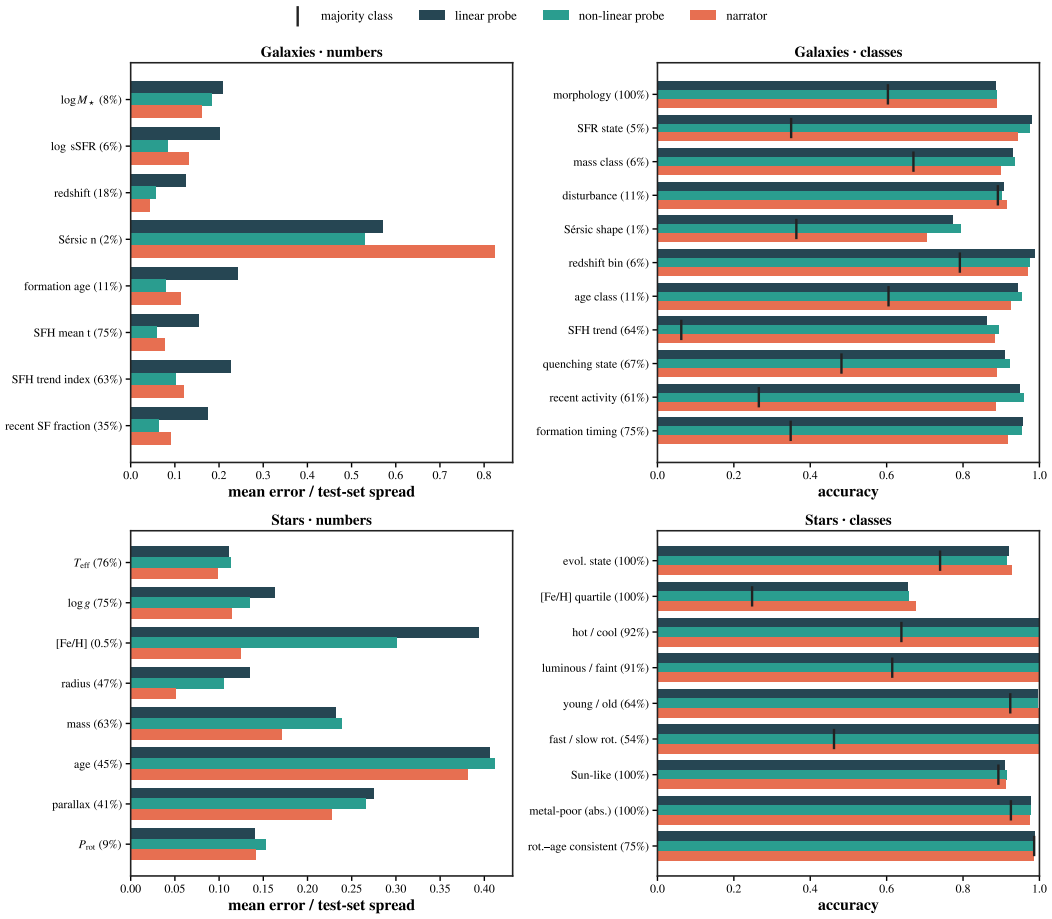


Figure 7: **Probes against the narrator, per property**, AstroSage-8B, single objects. Probes read the same tokens as the narrator. Left: mean absolute error divided by the spread (standard deviation) of the property in the test set; right: accuracy of class statements, with the rate of the most common class (black tick). All bars are on the objects where the narrator stated the property; the percentage next to each name is how often it did. Top: galaxies (3,000, none in the training text); bottom: stars (19,010, 99.2% described in the training text).